29/07/26 Activity-2

1. Stack:

push(54) → [54]

push(52) → [54, 52]

pop() = 52

push(55) → [54, 55]

push(62) → [54, 55, 62]

s = pop() = 62 , so s = 62.

Queue:

enqueue(21) → [21]

enqueue(24) → [21, 24]

dequeue() → removes 21

enqueue(28) → [24, 28]

enqueue(32) → [24, 28, 32]

q = dequeue() = 24.

s + q = 62 + 24 = 86

2. postfix expression

Expression: 10 5 + 60 6 / * 8 -

10 + 5 = 15

60 / 6 = 10

15 * 10 = 150

150 - 8 = 142

3. minimum no. of enqueue operation.

Q1 → 4 elements Q2 → 4 elements

4. Reverse operation

stack operation O(n)

5. postfix expression

8 2 3 ^ / 2 3 * + 5 1 * -

After first * is evaluted

The top two stack elements are 6, 5

6. abc*+def^n*

Infix to postfix